

Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Abstract

This document is a first-person reconstruction of the chain of thought for the original AI derivation of the closed-form tree amplitude A_n in the two-minus sector of one-dimensional deep-water surface waves. It is rebuilt from the visible run trajectory and tries to keep the original wording, the false starts, and the hedges rather than tidy them away.

Rewritten Chain of Thought

I can only use the prompt and `OnShellBG.m`, and I have to be careful not to read any other pre-existing file. I need a closed-form A_n for $\sigma = (-1, -1, +1, \dots, +1)$, and I should understand how the kernels work for a faster implementation. First question: is `wolframscript` even available? Let's run `wolframscript`. It is. I think about how to use the code: maybe I copy some definitions, maybe I append to `OnShellBG`, maybe I write a separate loader that does `Get`. I write a wrapper that builds the two-minus sign vector, calls `MakeKinematics`, and feeds `BGAmplitude`, then I just `Get` the file — and that was a mistake: `Get` runs the file's own demonstration tests, which are for a *different* sector ($\sigma = (-1, +1, \dots, +1)$, where the amplitude is identically zero), and it prints a wall of “ $A_5 = 0, A_6 = 0, \dots$ ” while the $n = 8$ case grinds. I let it run for a bit, decide it is wasting time, and interrupt it. Better idea: import the file as text, split it at the `VI. TESTS` marker, and `ToExpression` only the part before that, so I load the definitions without the demo. (A `Do[...,{n,fw},...]` trips a “Tag List is Protected” error too, so I switch the loop to `Scan`; small things.)

Now the oracle behaves. The first real surprise is $n = 4$: the recursion returns `Indeterminate` — “ $1/0$ ”, “ $0 \cdot \text{ComplexInfinity}$ ”. The on-shell point forces an internal two-leg current to zero momentum, so a propagator blows up and a real cancellation hides behind a removable $0/0$. I set $n = 4$ aside, to come back as a limit, and look at $n = 5, 6, 7$, which give clean, purely imaginary rationals like $-2304i$. So I write $A_n = iP_n$ and chase P_n .

I keep thinking about kernels, propagators, and the overall dimensions. Scaling $\omega \rightarrow \lambda\omega$ multiplies A_n by λ^{2n-4} ; changing g shows $A_n \propto g^{-(n-3)}$; and a family where the second negative leg goes soft, $\omega_2 = a \rightarrow 0$, shows the amplitude going like a^5 when ω_2 is the smallest scale around. Putting the degree, the g -power, and that soft behaviour together, the simplest thing that fits is the monomial

$$A_n \stackrel{?}{=} i \frac{2^{n-1}}{g^{n-3}} \omega_1 \omega_2^{2n-5}.$$

It is suspiciously clean, and it does reproduce the soft data. But it breaks the instant ω_2 is not the smallest same-sign scale. I notice something intriguing while poking at this: flipping ω_1, ω_2 both negative or both positive gives the same $B \equiv A/(-i)$, so whatever P_n is, it looks symmetric in the two negative legs. That tempts me to fit a symmetric polynomial $f(\omega_1, \omega_2)$ of degree six — $a(u^6 + v^6) + b(u^5v + uv^5) + c(u^4v^2 + u^2v^4) + d u^3v^3$. I solve for the coefficients on four points; the solver returns monstrous fractions, and they miss every other point by huge amounts. So A_n is simply not a polynomial in the two negative frequencies. I toy with a divided-difference / Lagrange form, $\sum_i \sigma_i \omega_i^{2n-4} \prod_{j \neq i} \omega_j / (\omega_i - \omega_j)$, and with B-splines on the sorted (negative) phase velocities, “normal-form coefficients based on the max of the frequencies” — these feel close to the right shape but I have no derivation for them yet.

The derivation is hiding in the absolute values. The only non-analytic thing in the whole construction is $|k_S| = |\sum_{i \in S} \sigma_i \omega_i^2|/g$ inside each propagator. An absolute value has a corner where its argument changes

sign, so A_n cannot be one polynomial — it has to be a *spline*, a piecewise polynomial with break surfaces $\sum_{i \in S} \sigma_i \omega_i^2 = 0$. That is exactly why my monomial held in one regime and failed in another: I had been sitting in one cell and then crossed a wall.

So I stop guessing and resolve the absolute values by hand, chamber by chamber. I take the symbolic $n = 5$ amplitude, collect every **Abs** argument, replace each one by $\pm$ itself according to its sign at a sample point in the cell, and simplify. Inside each cell the expression is an honest rational function, and after dividing out the common prefactor $i 2^4 \omega_1 \omega_2$ the pieces are, with $x = \omega_2$ and the positive legs y, z ,

$$x < y, z: x^4, \quad y < x < z: x^4 - (x^2 - y^2)^2, \quad y, z < x: x^4 - (x^2 - y^2)^2 - (x^2 - z^2)^2 + (x^2 - y^2 - z^2)^2.$$

These are not separate answers; they are the running inclusion–exclusion corrections of one object. The function “looks like $x^4 - \sum (x^2 - y_j^2)_+^2$ ”. Writing $t = \omega_2^2$, $a_j = \omega_j^2$, $m = n - 3$, and using the truncated power $[u]_+ = \max(u, 0)$, every cell is one value of

$$T_m(t; \{a_j\}_{j \in R}) = \sum_{S \subseteq R} (-1)^{|S|} \left[t - \sum_{j \in S} a_j \right]_+^m,$$

the truncated-power / B-spline form I had been circling. The walls $t = \sum_{j \in S} a_j$ are the $|k_S|$ sign changes, and below all of them only the $S = \emptyset$ term lives, so $T_m = t^m$ — which is precisely the soft-regime monomial I started from. The whole thing is one truncated power times a universal prefactor,

$$A_n = i \frac{2^{n-1}}{g^{n-3}} \omega_1 \omega_2 T_{n-3}(\omega_2^2; \{\omega_j^2\}_{j=3}^{n-1})$$

with $R = \{3, \dots, n-1\}$ the $n-3$ positive plus-legs.

Two loose ends. For $n = 4$ the spline has $m = 1$, so the formula predicts $A_4 = i(8/g) \omega_1 \omega_2 \min(\omega_2^2, \omega_3^2)$, and I recover the removable $0/0$ honestly as the $\varepsilon \rightarrow 0^+$ limit of the energy-conserving deformation $\omega = \{-\omega_3 - \varepsilon, \omega_2, \omega_3, -\omega_2 + \varepsilon\}$; the limit is finite and gives $-192i$ for $\{2, 3\}$, matching. The other loose end is which leg is “ t ”, and here I genuinely stumble: my first channel guess predicts -3776 where the oracle says -701 for one point, and it is fine for a small ω_2 but goes wrong once ω_2 is not small. I think the discrepancy is about signs, and I reframe it as a channel-selection question: counting left- and right-movers, my form works for one negative leg against $n-3$ same-sign plus legs, and the right rule is to let p be the $\sigma = -$ leg whose sign is shared by exactly $n-3$ of the plus legs (the set R), with q the other $\sigma = -$ leg, so $A_n = i 2^{n-1} q p T_{n-3}(p^2; \{\omega_j^2\}_{j \in R}) / g^{n-3}$.

My first script for this channel-aware form fails on every line — but the failure is a **Function::slotn** slot mistake in my selector code, not a disagreement with the oracle. I fix the selector and rerun, and now the closed form reproduces **BGAmplitude** *exactly*: the relative error is identically zero in exact rational arithmetic for $n = 4, 5, 6, 7$, at many points per n , including non-generic regimes where one frequency is a hundred times larger or smaller than the others, and at $g \neq 1$.

I want to be honest about the scope rather than overclaim. For arbitrary positive free frequencies handed to **MakeKinematics** — the setup the prompt actually specifies — energy conservation drives the two solved legs negative and the kinematics always land in this one physical channel; I check five hundred random points for each $n = 4, \dots, 8$ and the boxed formula is exact on all of them. If instead one hand-flips the signs of the free frequencies into other cells of the spline, the same truncated-power structure persists but with a different leg as p and a different same-sign set as R ; I state that selection rule but I did not fold every sign cell into a single universal expression. So, honestly: within the prompt’s own kinematics this is the closed-form answer and it is verified to exact equality; the fully sign-general statement I leave as the channel-selection rule plus the spline structure, not as one compressed formula.